Epistemology, Phase 2: After the Classical Canon

What to learn once Popper, Kuhn, Lakatos, Quine, Jaynes, Mayo and Pearl are behind you

Five frontiers: deciding under uncertainty • knowing together • reflexive systems • the physics of inference • the ethics of belief

The Shape of Phase 2

Phase 1 answered: *how should a single mind form and test beliefs about a stable world?* Every frontier beyond it relaxes one of those assumptions:

- What if you must **act**, not just believe? → **Decision theory & deep uncertainty**
- What if knowing is done by **groups**, not individuals? → **Social epistemology**
- What if the world **reacts to your beliefs about it**? → **Reflexivity & complex systems**
- What if inference is a **physical process** with costs and limits? → **Information, computation & inference**
- What if belief has **stakes for others**? → **Ethics of belief & epistemic risk**

Each track below gives the core problem, the key thinkers/readings, and why it bites for you specifically.

Frontier 1 — Decision Under Deep Uncertainty

Phase 1 quietly assumed you can always put a probability on things. The frontier question: what do you do when you *can't* — when even the hypothesis space is unknown?

Knight & Keynes — risk vs. uncertainty. Frank Knight's 1921 distinction: *risk* is measurable (known odds), *uncertainty* is not (unknown odds, unknown outcomes). Keynes' *Treatise on Probability* argues some probabilities aren't even comparable. This is the formal name for what regime change does to your backtests.

Savage & the axioms of rational choice — then the cracks. Savage's theorem derives expected-utility + Bayesianism from innocent-looking axioms (the high-water mark of 'rationality as theorem'). Then the **Ellsberg paradox**: people rationally-feeling violate the axioms when odds are ambiguous — leading to **imprecise probability** (credence intervals instead of points) and **ambiguity aversion** (Gilboa–Schmeidler maxmin expected utility). Ask yourself: should your position sizing use a prior, or a *set* of priors?

Kay & King — *Radical Uncertainty* (2020). Two heavyweight economists arguing that 'what is going on here?' beats probability puzzles for real decisions; direct attack on misused Bayesianism in finance and policy.

Small-world vs. large-world Bayes. Savage himself warned his theory only works in 'small worlds' where all possibilities are listed. Binmore's *Rational Decisions* develops the critique. Your Polymarket models are small worlds; markets are large ones — the gap is where blowups live.

Frontier 2 — Social Epistemology: Knowing Together

Phase 1's knower was alone. Real knowledge is produced by communities, markets, institutions — and prediction markets are literally social epistemology machines.

Testimony & expertise. Most of what you know, you know because someone told you. When is that rational? How do laypeople identify experts without being experts (Goldman's 'Experts: Which Ones Should You Trust?')? Directly relevant to sell-side research consumption — your day job's product *is* testimony.

Peer disagreement. If an epistemic peer — same evidence, same competence — disagrees with you, should you move? Conciliationists say yes (split the difference), steadfasters say no. Every trade is a bet that a peer (the counterparty) is wrong: trading is applied disagreement theory.

Aggregation mechanisms. Condorcet's jury theorem, wisdom-of-crowds conditions and failure modes (correlated errors, cascades), Hayek's 'The Use of Knowledge in Society' (prices as distributed computation), and the epistemology of prediction markets — when do they aggregate information vs. amplify herding? Your whale-tracking module is an implicit theory of exactly this: whose signal is information, whose is noise.

Epistemic pathologies of institutions. Groupthink, information cascades, audit cultures, Goodhart's law as epistemology (when a measure becomes a target it stops measuring). Ties to your political-economy writing: institutions as belief-forming machines that can be well or badly designed.

Frontier 3 — Reflexivity: When the World Reads Your Model

Phase 1's world sat still while you studied it. Markets don't: beliefs about the system are inputs to the system. This is the deepest structural difference between physics and finance — and classical philosophy of science mostly ignores it.

Soros — reflexivity. Dismissed by academics for years, now taken seriously: participants' biased views change fundamentals, which validate the views (bubbles as epistemic feedback loops). *The Alchemy of Finance*, filtered through your Popper training (Soros was Popper's student — reflexivity is his answer to why falsification fails in markets).

Self-fulfilling and self-defeating prophecies as a formal problem. Merton's original essay; then the modern version — **performativity** (MacKenzie's *An Engine, Not a Camera*: how Black-Scholes didn't describe option markets but *created* them). The strongest available case that the map rewrites the territory.

Alpha decay as an epistemological law. In physics, publishing a law doesn't repeal it; in markets, discovering an edge destroys it (Lo's Adaptive Markets Hypothesis is the framework here). Induction in a reflexive system has a shelf life *by construction* — this deserves to be thought through philosophically, and almost nobody has.

Complex systems epistemology. Emergence, non-ergodicity (Ole Peters' ergodicity economics — time averages vs. ensemble averages, with direct position-sizing implications), and the limits of prediction in systems with feedback. When is a system knowable at all?

Frontier 4 — The Physics of Inference

Phase 1 treated inference as free and instant. It isn't: it's computation, done by bounded physical systems, and the constraints are themselves epistemology.

Information theory as epistemology. Shannon entropy, KL divergence as the currency of belief-updating, maximum entropy as the honest prior (Jaynes' other big idea — a natural sequel to his book). Minimum description length (Rissanen): Occam's razor as compression, the practical cousin of Solomonoff.

Bounded rationality. Simon's satisficing; Gigerenzer's ecological rationality — when do fast heuristics *beat* optimization (less-is-more effects)? The rehabilitation of rules-of-thumb as rational under computation limits, against the 'biases' literature (Kahneman) that treats them as defects. Read both sides.

Computational complexity of inference. Exact Bayesian updating is NP-hard in general — the ideal reasoner of Phase 1 is physically impossible. What does 'rational' mean for an agent that can't afford coherence? (Aaronson's essay 'Why Philosophers Should Care About Computational Complexity' is the fun gateway.)

Predictive processing & the free energy principle (optional, speculative). Friston, Clark's *Surfing Uncertainty*: the brain as a hierarchical Bayesian machine minimizing prediction error. Contested, occasionally unfalsifiable-feeling (apply your Popper!), but it's the most ambitious attempt to make epistemology a natural science — and it's structurally a Kalman filter all the way down, which you'll appreciate.

Frontier 5 — The Ethics of Belief & Epistemic Risk

Phase 1 asked what's rational to believe. The last frontier asks what's *responsible* to believe — because beliefs have consequences for other people, and error costs are asymmetric.

Clifford vs. James — the founding duel. Clifford: 'it is wrong, always, everywhere, to believe on insufficient evidence' (the shipowner parable). James' 'The Will to Believe': sometimes you must decide before the evidence is in. Short, classic, both free online.

Inductive risk. Rudner/Douglas: since evidence never *proves*, accepting a hypothesis always risks error — and choosing the confidence threshold requires weighing the *costs* of each error type. Science can't be value-free even in principle. (You already live this: a false-positive edge costs money, a false-negative costs opportunity — your significance threshold IS a value judgment.)

Epistemic injustice (Fricker). Whose testimony gets discounted, whose concepts are missing — knowledge-production as a site of fairness. Connects to your commons/liberalism writing: epistemic commons, who gets to contribute to shared knowledge.

Pragmatism as the umbrella. Peirce (who anticipated *everything* — fallibilism, abduction, truth as end-of-inquiry), James, Dewey; modern heirs like Rorty (radical) and Misak (sober). Pragmatism is arguably where your entire conversation was heading: truth as what survives inquiry indefinitely, belief as a rule for action. Peirce's 'The Fixation of Belief' is 15 pages and reads like it was written for you.

How to Sequence Phase 2

Don't take these in order — take them by resonance. But if you want a default path:

Step	Read first	Why this order
1	Peirce, 'The Fixation of Belief' + Clifford & James	Short classics; reframes everything from Phase 1 as pragmatism
2	Knight/Keynes distinction + Ellsberg + imprecise probability (SEP)	Breaks the 'everything has a probability' habit before it hardens
3	Hayek 'Use of Knowledge' + peer disagreement (SEP) + jury theorems	Social epistemology via the market door — your native entrance
4	Soros + MacKenzie + Peters (ergodicity)	Reflexivity: the finance-specific frontier almost no one works on
5	MaxEnt/MDL + Gigerenzer vs. Kahneman + Aaronson essay	Inference as bounded computation; sharpens all your modeling
6	Inductive risk (Douglas) + Fricker	Closes the loop: thresholds as values, knowledge as institution

The through-line: Phase 1 taught you that knowledge is fallible testing. Phase 2 teaches that the tester is embedded — in a decision, in a crowd, in a feedback loop, in a physical body, in a web of obligations. The frontier where *you* could actually contribute something original: **the epistemology of reflexive systems** — Popperian severity + ergodicity + performativity, applied to markets. It's underworked, it needs someone with both the philosophy and the P&L, and it's a natural spine for the Substack.